MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED SCHOTTKY TTL,
NAND GATES, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/3308, dated 5 April 1983, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title, delete and substitute as printed above.

* 1.2.3, delete case outlines X and Y and respective descriptions, and substitute:

"2 C-2 (20-terminal, .350" x .350"), square chip carrier package
X C-2A (20-terminal, .350" x .350"), square chip carrier package"

* 1.3, Thermal resistance, junction-to-case (θ_{JC}): Delete and substitute as follows:

"Cases A, B, C, D, X and 2- - - - (See MIL-M-38510, appendix C)

* 1.3, Junction temperature (T_J), add "2/" after (T_J).

* After footnote 1/ add the following footnotes:

"2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883."

PAGE 2

1.4, after Supply voltage, add "(V_{CC})".

1.4, delete "Normalized fanout (each output) 2/:" and substitute "Normalized fanout (each output) 4/:".

3.2.3, delete and substitute:

"3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request."

Bottom of page, delete footnote "2/" and substitute footnote "4/".

PAGE 3

* TABLE I, I_{IL}, min limits column, delete "-.03" and substitute "0".

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PAGE 4

* TABLE II, group A test requirements, class B devices, add "10, 11".

TABLE II, Interim electrical parameters, delete "(pre burn-in)".

* TABLE II, delete "Additional electrical requirements for group C periodic inspections" in its entirety.

3.6, Marking, delete second sentence.

* 4.2a(1): Delete "Test condition D or E" and substitute "Test condition D, E, or F".

PAGE 5

4.2c, delete and substitute: "c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510."

* 4.4.3b, delete in its entirety.

4.4.3c(1): Delete "D or E" and substitute "D, E, or F".

4.4.3c(3): Delete "appendix B of MIL-M-38510." and substitute "method 1005 of MIL-STD-883."

PAGES 7, 8, 9, AND 10

* FIGURE 1: Delete "Case X, Y" and substitute "Case X, 2".

PAGES 12, 13, 14, AND 15

* FIGURE 3, line 2, delete "and Y" and substitute "and 2".

* FIGURE 3, line 3, delete "for cases X and Y" and substitute "for cases X and 2".

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* FIGURE 3: Add optional burn-in circuit as printed on page 4 of this amendment.

PAGES 17, 18, 19, 20, 21, 22, 23, AND 24

* TABLE III, column four: Delete "Cases X, Y" and substitute "Cases X and 2".

PAGES 18, 20, 22, AND 24

TABLE III, Subgroup 9, t_{PHL}, (device type 01, test nos. 55-62), (device type 02, test nos. 51-56), (device type 03, test nos. 57-65), (device type 04, test nos. 49-56), minimum limits column: Delete "2.0" and substitute "1.5".

PAGE 18

* TABLE III, device type 01: Delete note 2/ and substitute:

"2/ I_{IL} limits shall be as follows:

Test	Min/max limits in mA			
	Circuits			
	A	B	C	D
I_{IL}	-.25/-.60	-.03/-.60	-.03/-.60	0/-0.3

"

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PAGE 20

* TABLE III, device type 02: Delete note 2/ and substitute:

"2/ I_{IL} limits shall be as follows:

Test	Min/max limits in mA			
	Circuits			
	A	B	C	D
I_{IL}	-.25/-.60	-.03/-.60	-.03/-.60	0/-0.3

"

PAGE 24

* TABLE III, device type 04: Delete note 2/ and substitute:

"2/ I_{IL} limits shall be as follows:

Test	Min/max limits in mA			
	Circuits			
	A	B	C	D
I_{IL}	-.25/-.60	-.03/-.60	-.03/-.60	0/-0.3

"

PAGE 25

6.1, delete in its entirety.

6.4, delete "Electrical ground (common terminal)" and substitute "Ground zero voltage potential".

PAGE 26

* TABLE IV, delete and substitute:

"TABLE IV. Manufacturers' designations.

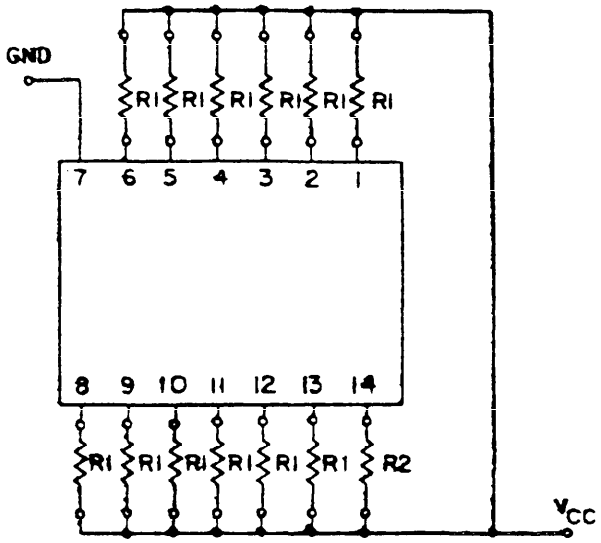
Device type	Manufacturer			
	Circuit A	Circuit B	Circuit C	Circuit D
	Fairchild Semiconductor	Motorola Incorporated	Signetics Corporation	Texas Instruments
01	X	X	X	X
02	X	X	X	X
03	X	X	X	
04	X	X	X	X

"

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Device types 01, 02, 03, and 04
Excluding case 2

High-temperature (175°C) burn-in (optional circuit)



NOTES:

1. High temperature (175°C) condition F burn-in circuit only.
2. R1 = 2k Ω, R2 = 2Ω.
3. V_{CC} shall be chosen such that 5.0 V minimum is present at device terminal.

FIGURE 3. Burn-in and life test circuit - Continued.

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Navy - EC

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Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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